My Graduate School Application Experience

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UF Class of 2020

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This document details my graduate school experience in case it is of any help to any of you in the coming application process for an Astronomy Ph.D. program.

I'll start first with some information about me that I submitted to graduate schools and other resources that may help you in this journey. I will then discuss some things about the process after the preliminary information.

My Qualifications:

Academic:

- Cumulative GPA: 3.65/4.00
- Degree: Dual Major in Astrophysics (BS) and Physics (BS)
- GRE Scores (Score/Percentile):
    - 1st Time: Verbal 154/64% Math 154/53% Writing 4.0/57%
    - 2nd Time: Verbal 160/86% Math 156/60% Writing 4.5/81%
- PGRE Scores (Score/Percentile):
    - 1st Time: 560/19%
    - 2nd Time: 560/19% (Yes they are terrible… I know)
- Dean's List Recipient for multiple semesters

Research:

- Worked on and produced important results for two major projects in both observational and theoretical/simulation astronomy.

Outreach:

- Quite minimal:
    - Starry Night (2017, 2018, 2019)
    - Astronomy & Astrophysics Society (Member and Treasurer)

Rec. Letters:

- 2 UF faculty, 1 external
    - All 3 three well known in their respective field

Skills:

- Comp. Languages: Python, Matlab, C, bash, tcsh
- OS: Windows, Linux, ChromeOS
- Obs. Skills: Experienced in taking observations using 14" and 12" Meade telescopes, using CCD's in the optical range, and writing code to reduce images and analyze images.

Schools Applied to and some info on them in 2019:

School	PGRE	GRE	Deadline	App Fee ($)	Submitted ?		Waived Fee
University of Arizona	O	Y	12/09	85	Y	O	Opt Score Report
UNC	N	Y	12/10	95	Y		Accepted
Ohio State	O	Y	12/13		Y		Rejected
University of Maryland	Y	Y	12/13	75	Y		Waitlisted
Harvard	O	N	12/15	105	Y		Sent Score
Cornell	O	N	12/15	80	Y		
UT Austin	N	Y	12/15	65	Y		
University of Michigan	N	Y	12/15		Y		
University of Wisconsin (M)	O	Y	12/31		Y		
University of Florida	O	Y	01/08	30	Y		
University of Illinois	O	Y	01/15		Y		
NMSU*	N	N	01/15	40	Y		
UMASS Amherst	O	Y	01/15	80	Y		

* I will explain later the NMSU decision.

Cost of me Applying:

Item	Cost ($)
PGRE 1*	75
GRE 1*	102.5
PGRE 2*	75
GRE 2*	102.5
Transcript to Michigan	6
Transcript to Ohio St.	6
Transcript for me	6
Sending GRE Scores**	189
UT Austin App fee	65
UF App fee	30
NMSU App fee	40
Total	697

* I qualified for a reduced fee of 50% normal test price.
** Everyone gets 4 free scores to send and the rest are $27 each.

Copy of my full CV and Personal Statement:
CV:
https://drive.google.com/file/d/1Kv6krhrA4KHQDMNl5OkYTEuXNWBZ-l0Y/view?usp=sharing

Personal Statement / Statement of Purpose
https://drive.google.com/file/d/1zVlEDJnFXLwOMSvSRLzxRI75Rb4ImSBs/view?usp=sharing

Helpful P.S Link:
https://twitter.com/astrocaits/status/1201157164489986048

Interview: Possible Q's

Questions provided by: Dr. Paul Torrey

1) Tell us about your research

2) Why do you want to come to this university?

3) Describe a research project you worked on and discuss what you liked most as well as what you found most difficult.

4) Describe an activity, academic or not, in which you took a leadership position.

5) Discuss a class or research project in which you worked with other students and how you dealt with difficulties you encountered.

6) Discuss your programming experience and how comfortable you are with various languages.

7) Talk about which of your skills/qualities will make you successful in grad school and what you think you still need to work on.

8) List professors at the school who you are interested in working with and, more generally, ideas for projects and directions you find interesting.

9) Talk about your interests/hobbies that are not related to research or academics.

Other link:
https://sites.google.com/site/taurusastrout/graduate-school/interviewing

Discussion:

Like any advice you receive for graduate school you should take what I have to say with a grain of salt. However, if enough peers, professors, and other people or mentors tell you enough of the same thing, then you should probably weigh those pieces of advice more. You may be feeling nervous as you are getting ready to apply as I was and that is natural. Am I good enough? Why would they take me? Can I afford to apply? These are some of the questions that may be going through your head and I'll try to touch on them as best I can.

What schools to apply to?

Personally I was quite worried about taking the PGRE and GRE because of how bad I am at standardized testing so when choosing schools I went by this: https://docs.google.com/spreadsheets/d/19UhYToXOPZkZ3CM469ru3Uwk4584CmzZyAVVwQJJcyc/edit?usp=sharing
Whether or not you are worried about testing this list is still a good resource since it compiles all the Physics and Astronomy schools with Phd programs in one place so start looking there. Check out their websites, what research projects they have, and maybe even contact some people there. I crossed off and added schools just by sending any email to one of the department heads asking if they had the topics of research I was interested in. Also if you are confused about anything in their application, definitely send an email or call.

Another piece of advice from me, is don't be narrow minded in picking a place because of location or overly specific research projects. While you may have an idea of what you want to work in, it may change and turns out you like something else. What I recommend is to talk to your mentors and advisors and get their input on specific departments; that may help you narrow down choices. As for location, if you absolutely hate a specific type of climate then in that case you can be more narrow minded (you definitely do not want to be miserable for 5 years) but if you are not adamant about hating the snow for example then do not worry about location too much.

Costs

Applying to graduate school can get ridiculously expensive and whether or not GRE's and application costs are a total scam is an entire topic by itself. However, there are several ways you can cut costs. Numerous universities have application fee waivers you may qualify for. ETS (company that makes the GRE and its subject tests) provides a 50% discount on exams if you demonstrate financial need. Personally I qualified for several application fee waivers and the GRE waivers, saving me hundreds of dollars. What you cannot waive are the costs for sending your scores. They are $27 each but you do get 4 free ones so pick wisely. By that I mean look into the schools that even require it and make those a priority. It is quite unfortunate how expensive this process is but at least there are some ways you can cut costs. Also if you are applying to these waivers, prepare any materials well in advance, I missed out on a waiver because I waited too long to contact the people I needed to.

Interviews

Not all schools do interviews but if they do and they invite you to one then jump on that opportunity because it means you're on their shortlist. I gave a list of possible questions that my advisor gave to me on what to expect and he also provided the link in that section. I recommend if you get an interview to go through these questions and prepare an answer since those questions are quite common and I was asked a few of them. Another common question is "What do you hope to get out of your graduate school experience?". Here they are essentially testing you if you have thought long and hard about going. Another question is about your research and it was "Why does your research matter?". They are not insulting you with this question at all, they want to know if you have actually been involved in the inquisitive aspect of research and know what the direction of the research is and why you are doing it rather than you just running code and showing the results to your advisor mindlessly.

I did not get in anywhere directly, I had an interview for the two places I got in and for NMSU they told me I was in the top 10% but still wanted an interview but I rejected them because I was no longer interested in their department because of my other options. Besides being nervous I also just tried to be myself during the interview while also being professional. I made my interviewers laugh and we had an enjoyable time (as enjoyable as an interview can be). I say this not because you should try to be funny or anything like that (that could backfire) but when you are being interviewed, not only are you being tested on your proficiency in research but they also want someone who they think will fit in their department not just professionally but personally. It helps

them make a decision if they think a candidate will mostly get along with everyone and shows promising research/academic skills.

Test Scores:

Of course I do not sit on an admissions committee so I do not know how much these weigh and it will definitely vary between institutions but I can give some personal insight to these exams.

The PGRE was really tough for me and in the end it was pointless. I only had one school that required it and the rest were either optional or not required. So please look into your list of schools and think about whether or not you really need to take this exam. If you find several of your schools require it or you just feel like you should take it then I'll give you a little advice (I say a little because I performed terribly on it). I will start off by saying start prepping early as in several months in advance. There are a few old exams available online so I recommend taking one cold with no studying and then see what topics you got wrong. If your answer is, I got everything wrong and need to study everything then if you took this "diagnostic exam" months in advance then this is where you made the right choice because now you can go topic by topic and learn as much as possible. The sections with most weight are Mechanics and EM so if you wish to do well, make sure you know these topics as best as possible. The rest of the topics try to learn but if you are pressed for study time then pick the next two highest weight sections and focus on those. If you only did bad on a few topics then first off, I commend you for being so studious in physics classes and two, just focus on those topics and brush up on the topics you did well on slightly just so nothing surprises you. Most importantly good luck! The test is long (100 Q's) and super exhausting, I found myself wanting to just give up by question 70.

The GRE is more manageable. I have been told to make sure to do well in the math section and well I did not do too well but I still got in some places. Most schools will require it so make sure to study well for it since it is very expensive ($205 full price). I will attach a link to a book of practice problems you can use to study from. Especially for the math you want to get through the problems as quickly as possible since they are not hard but can be tricky and require you to use methods last employed in the 7th grade. I have no advice for the verbal and writing unfortunately besides making sure you can type quickly when writing your essays. To be honest I have no clue how I improved my verbal score the second time I took the test considering I did not study for the verbal or writing at all.

https://drive.google.com/open?id=17oUpxkyfsIudmwXyPYH-aUrCx9Bdzq7I

What can I do now to improve my chances?

Once again I do not sit on an admissions committee but I'll give some advice on how to at least prepare yourself for the process and some more general advice to the sophomores and juniors.

- Start early. This means begin making your school list well before the fall semester, inquiring whether you qualify for any waivers, and emailing professors you are interested in working with to see if they are taking grad students. If you can also make your fall semester when you are applying as light as possible that would be ideal. You may find that there are still things for applications that need to be done and you do not want to be dealing with hard classes while applying and potentially hurting your applications by rushing them.
- Have your personal statement proofread and critiqued by as many people as possible, especially professors who are committee members in your local department that have more insight on what schools are looking for. I linked a helpful Twitter thread originally emailed to me and other undergraduates by Dr. Desika Narayanan. The thread is by Dr. Caitlyn Cassey from UT Austin and her advice really helped write my own personal statement along with input from my advisors. Similar advice goes for your CV, write what skills and qualifications you want to highlight and have it proofread by everyone. I have attached my own CV as an example of what it can look like if you are stumped.
- Having outreach experience is great however depending on who you ask they will give a different answer as to how important it is. Some say it makes up an entire pillar of your application (the others being grades, research and test scores), others will say it is merely a tie-breaker when committee members are making their final candidate list. My advice is that it does not hurt having some outreach experience, and it benefits the general community too!
- For those of you still a year or two or more away from applying I want to make a note on research. In the beginning you will more than likely just be working with data reduction pipelines or something basic. This is perfectly fine because that is how everyone starts and I did it too. However I want to note that you should not be stuck just doing data reduction. After you get the handle of things either your advisor will or you need to ask them to give you a more complicated task that may involve developing your own code, reading through the literature for information on what you are doing, and/or thinking through and working out complicated problems. I mention this because it is much preferred a candidate has experience with more complicated projects like they will do in grad school than just data reduction work. Also do not be afraid to switch research projects because it is good to explore. What you should always do though is see your

current project/task to its entirety like if your goal was to measure magnetic activity for neutron stars in a given dataset, make sure you complete working on that data set before you move on.

Closing Remarks:

At the end of the day you cannot guess what the result of these applications will be because it depends on how things proceed in these grad school committees. Normally (from what I have been told) there is a list of candidates made during the committee discussions and however the list looked like when they say they are done is in what order they will seek out their candidates. In that list you will have about three tiers of candidates where the first tier gets the offers first and then they'll go down the list as offers get rejected so you want to give them an application good enough that you will be in the top two tiers.

If you do not get in anywhere or the places you wanted to, maybe my next words will be encouraging. This process is quite random, you do not really know what to expect even though the committees are trying to be as fair and objective (hopefully) as possible. Your application could have been the first one where everyone was full of energy or it could be the one before lunch where people are starting to get hungry and want to speed things up or even at the end of the day where they just want to go home. What I am getting at is do not put yourself down if you are not happy with the results because it is not a reflection of you as a candidate or person but of how competitive the applicant pool was or where your application was in the giant stack of applications and who read your application. There are a lot of unknowns in getting to graduate school and there definitely is not a formula to get in, but you can do your best to increase your chances by making your application as good as possible. I got into one of the best Astronomy PhD programs in the country (Ohio State) but did not get into other places not as prestigious. I do credit my interview with helping my chances, in fact I wish more places would conduct interviews, but also me not getting into other places shows how unpredictable the process is.

I sincerely wish you the best and hope you get in somewhere that will make you happy and allow you to grow as an individual and professional. If you receive multiple offers then make sure to visit and ask many questions to faculty and graduate students to get a real feel for the place. You will be spending several years of your life in graduate school so pick wisely. Good luck!